Insurance Claim Analysis and Fraud Insights

Dataset Overview

Dataset Title: insurance_claims.csv

Source: https://data.mendeley.com/datasets/992mh7dk9y/2

This dataset contains detailed records of insurance claims including policyholder information, incident details, vehicle information, and claim amounts. Each row represents one insurance claim filed by a customer.

The dataset consists of 1000 records and 40 attributes. One of the key fields in this dataset is **fraud_reported**, which indicates whether the claim was identified as fraudulent (Y) or not fraudulent (N).

This project focuses on analysing insurance claims data using Excel and Power BI to identify patterns, trends, and fraud-related insights.

Attribute details

Attribute	Description
months_as_customer	Duration of customer relationship (in months)
age	Age of insured person
policy_state	State where policy was issued
policy_deductable	Deductible amount
policy_annual_premium	Annual premium amount
umbrella_limit	Additional coverage limit
insured_sex	Gender of insured
insured_education_level	Education level
insured_occupation	Occupation
incident_date	Date of incident
incident_type	Type of accident/incident
collision_type	Type of collision
incident_severity	Severity of incident
authorities_contacted	Authorities contacted
incident_state	State of incident
incident_city	City of incident
incident_hour_of_the_day	Time of incident
number_of_vehicles_involved	Vehicles involved in accident
property_damage	Property damage status
bodily_injuries	Number of injuries
witnesses	Number of witnesses
police_report_available	Police report availability
total_claim_amount	Total amount claimed
injury_claim	Injury claim amount

property_claim	Property claim amount
vehicle_claim	Vehicle claim amount
auto_make	Vehicle brand
auto_model	Vehicle model
auto_year	Manufacturing year
fraud_reported	Fraud status (Y/N)

Project Overview

This project involves analyzing insurance claims data to understand fraud patterns, claim trends, and customer behavior using Excel and Power BI tools.

1. Data Acquisition:

The dataset was obtained in CSV format and imported into Microsoft Excel for cleaning and preprocessing.

2. Data Cleaning (Excel):

- Handled missing values and special characters.
- Checked and removed duplicate records.
- Formatted date columns.
- Standardized categorical values.
- Verified numerical data consistency.

3. Data Analysis (Excel):

- **Created Pivot Tables to analyze:**
  - Fraud vs Non-Fraud distribution
  - Claim amount by incident type
  - Claim amount by state
  - Incident severity vs total claim amount
- **Used formulas such as:**
  - IF()
  - COUNT()
  - SUMIF()
  - AVERAGE()
- Created Pivot Charts for visual representation.

Problem Statement

Insurance companies process thousands of claims every year. Identifying patterns in claims and detecting unusual trends manually is difficult and time-consuming.

The dataset contains raw claim records with multiple attributes related to customers, policies, and incidents. However, without proper analysis, meaningful insights cannot be extracted.

The objective of this project is to clean and analyze insurance claims data using Excel and Power BI, identify fraud-related patterns, and create an interactive dashboard that helps in understanding claim behaviour and risk trends.

Tools & Technologies

Excel: Data cleaning, transformation, and Pivot Tables

Power BI: Data modelling, DAX calculations, visualization and dashboard creation.

Insights from Excel Dashboard:

1. Fraud cases represent a smaller portion of total claims but contribute significantly to total claim amount.
2. High severity incidents result in higher average claim amounts.
3. Certain collision types show relatively higher fraud occurrences.
4. Claims without police reports tend to have higher fraud frequency.
5. Some states show higher claim volumes, indicating higher insurance activity.
6. Monthly trends indicate fluctuations in claim frequency across different periods.

Types of Analysis Performed

Descriptive Analysis

Descriptive analysis was performed to summarize and understand the basic characteristics of the insurance claims dataset. Pivot Tables, charts, and KPI cards were used to calculate totals, averages, percentages, and distributions.

Examples:

- Total number of claims
- Total fraud and non-fraud cases
- Total and average claim amount
- Claim distribution by state
- Claim amount by incident type

Diagnostic Analysis

Diagnostic analysis was conducted to identify the reasons and patterns behind fraudulent claims. This involved comparing fraud cases with various attributes such as incident type, severity, police report availability, and claim amounts.

Examples:

- Fraud cases based on incident severity
- Fraud occurrence when police report is unavailable
- Relationship between claim amount and fraud status

This analysis helped in identifying factors associated with higher fraud risk.

Pivot table and Slicer

Fraud vs Non-Fraud

This pivot table shows the count of fraudulent and non-fraudulent claims. It helps in understanding the overall fraud distribution in the dataset.

Fraud vs Non fraud	Count of policy_number
N	753
Y	247
Grand Total	**1000**

Total Claim Amount by Incident Severity

This pivot analyzes the total and average claim amount based on incident severity. It helps identify how severity impacts claim value.

severity	Sum of total claim	Average of total claim
Major Damage	$1,76,82,540.00	$64,067.17
Minor Damage	$1,72,19,510.00	$48,642.68
Total Loss	$1,73,82,740.00	$62,081.21
Trivial Damage	$4,77,150.00	$5,301.67
Grand Total	**$5,27,61,940.00**	**$52,761.94**

Fraud by Collision Type

This pivot compares fraud cases across different collision types. It helps in identifying which collision categories are more fraud-prone.

Policy number count	Fraud reported		
Collision type	N	Y	Grand Total
Front Collision	184	70	254
No Collision	162	16	178
Rear Collision	201	91	292
Side Collision	206	70	276
Grand Total	**753**	**247**	**1000**

Claims by State

This pivot table displays the number of claims and total claim amount by state. It helps analyze geographic distribution of insurance claims.

State	Count of total claim	Sum of total_claim_amount
NC	110	$ 54,21,230
NY	262	$ 1,47,73,770
OH	23	$ 11,44,720
PA	30	$ 14,87,150
SC	248	$ 1,35,38,150
VA	110	$ 54,99,770
WV	217	$ 1,08,97,150
Grand Total	**1000**	**$ 5,27,61,940**

Fraud vs Police Report Availability

This pivot compares fraud occurrences based on police report availability. It helps determine whether missing police reports are linked to higher fraud.

Police report	Fraud reported
No	343
Unknown	343
Yes	314
Grand Total	**1000**

Incident Type Distribution

This pivot shows the frequency of each incident type. It helps understand which types of accidents are most common.

State wise								
Incident type	NC	NY	OH	PA	SC	VA	WV	Grand Total
Multi-Vehicle Collision	43	119	12	15	103	41	86	419
Parked Car	10	16	1	4	22	11	20	84
Single Vehicle Collision	43	112	7	9	105	44	83	403
Vehicle Theft	14	15	3	2	18	14	28	94
Grand Total	**110**	**262**	**23**	**30**	**248**	**110**	**217**	**1000**

Monthly Claim Trend

This pivot table displays claims based on month of incident. It helps analyze seasonal patterns in claim frequency.

Month	Total claim
Jan	516
Feb	472
Mar	12
Grand Total	**1000**

State Vs Policy premium

This pivot table displays the policy premium based on policy state. It helps to analyze how the patterns change for policy states.

State	Policy premium
IL	$ 4,23,559.80
IN	$ 3,89,172.29
OH	$ 4,43,674.06
Grand Total	**$ 12,56,406.15**

Claim Band Analysis

This pivot analyzes total claim amount based on claim size categories (Low, Medium, High). It helps identify which claim band contributes most to total claim payout.

Claim Band	Total claim amount
High Claim	$ 1,91,64,640
Low Claim	$ 97,830
Medium Claim	$ 2,98,410
Grand Total	$ 1,95,60,880

Age Band vs Fraud

This pivot compares fraud occurrence across different age groups. It helps identify which age segment has higher fraud probability.

Count of Claim	Fraud reported		
Age band	N	Y	Grand Total
Adult	122	37	159
Middle Age	88	37	125
Senior	7	2	9
Young	6	5	11
Grand Total	223	81	304

Slicer Usage

Slicers were added for Incident Type, State, Severity, and Fraud Status, Claim_Band,Customer_Segment to make the dashboard interactive. They allow users to filter all pivot tables simultaneously for dynamic analysis.

DAX Calculations

Age Band

The Age Band calculated column was created to categorize policyholders into demographic segments based on their age. This segmentation helps in analyzing fraud trends, claim behavior, and risk patterns across different age groups.

Structure | Formatting | Properties | Sort | Groups | Relationships | Calculations

```
Age Band =
SWITCH(
    TRUE(),
    'Cleaned dataset'[age] < 25, "Young",
   'Cleaned dataset'[age] <= 40, "Adult",
    'Cleaned dataset'[age] <= 60, "Middle Age",
    "Senior"
)
```

juries	witnesses	police_report_available	total_claim_amount	injury_claim	property_claim	vehicle_claim	auto_make	auto_model	auto_year	fraud_reported	Age Band
2	2	Yes	51590	9380	9380	32830	Audi	A5	2015	N	Adult
1	1	Yes	27700	2770	2770	22160	Toyota	Camry	2012	N	Adult
2	2	No	91650	14100	14100	63450	Accura	TI	2011	N	Middle Age
0	1	Yes	64800	10800	5400	48600	Audi	A3	1999	N	Adult
1	1	Yes	5330	1230	820	3280	Suburu	Legacy	2001	N	Middle Age

Claim Band

The Claim Band calculated column classifies total claim amounts into Low, Medium, and High categories using DAX logic. This enables aggregated analysis of claim distribution, risk exposure, and fraud patterns across claim size groups.

Structure | Relationships | Calculations | Calendars

```
Claim Band =
SWITCH(
    TRUE(),
    'Cleaned dataset'[total_claim_amount] < 5000, "Low Claim",
    'Cleaned dataset'[total_claim_amount] < 20000, "Medium Claim",
    "High Claim"
)
```

vailable	total_claim_amount	injury_claim	property_claim	vehicle_claim	auto_make	auto_model	auto_year	fraud_reported	Incident month	Age Band	C

Calendar Table

The Calendar table was created using DAX to generate a date range based on the incident dates. Year and Month columns were added to support time-based reporting. This table helps in performing YTD and year-over-year analysis in the dashboard.

Structure | Manage relationships (Relationships) | New measure, Quick measure, New column (Calculations) | New table, Mark as date table (Calendars)

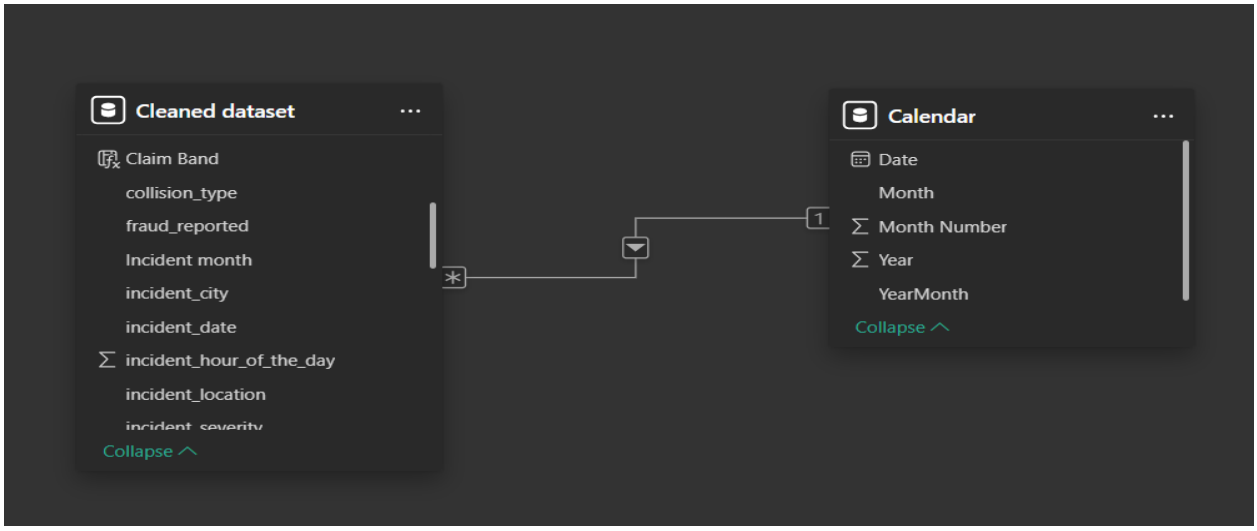
```
Calendar =
ADDCOLUMNS(
    CALENDAR(
        MIN('Cleaned dataset'[incident_date]),
        MAX('Cleaned dataset'[incident_date])
    ),
    "Year", YEAR([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Month Number", MONTH([Date]),
    "YearMonth", FORMAT([Date], "YYYY-MM")
)
```

Date	Year	Month	Month Number	YearMonth
01-01-2015 00:00:00	2015	Jan	1	2015-01
02-01-2015 00:00:00	2015	Jan	1	2015-01
03-01-2015 00:00:00	2015	Jan	1	2015-01
04-01-2015 00:00:00	2015	Jan	1	2015-01
05-01-2015 00:00:00	2015	Jan	1	2015-01
06-01-2015 00:00:00	2015	Jan	1	2015-01
07-01-2015 00:00:00	2015	Jan	1	2015-01
08-01-2015 00:00:00	2015	Jan	1	2015-01
09-01-2015 00:00:00	2015	Jan	1	2015-01
10-01-2015 00:00:00	2015	Jan	1	2015-01
11-01-2015 00:00:00	2015	Jan	1	2015-01
12-01-2015 00:00:00	2015	Jan	1	2015-01
13-01-2015 00:00:00	2015	Jan	1	2015-01
14-01-2015 00:00:00	2015	Jan	1	2015-01
15-01-2015 00:00:00	2015	Jan	1	2015-01
16-01-2015 00:00:00	2015	Jan	1	2015-01
17-01-2015 00:00:00	2015	Jan	1	2015-01
18-01-2015 00:00:00	2015	Jan	1	2015-01
19-01-2015 00:00:00	2015	Jan	1	2015-01

A **star schema model** was implemented by connecting the Calendar dimension table to the insurance_claims fact table. The Calendar table stores date-related attributes, while the insurance_claims table contains transactional claim data. This structure improves model performance, scalability, and enables accurate time-based analysis.

Data Modelling

A relationship was created between the incident date in the claims table and the Date column in the Calendar table. It is a many-to-one relationship because many claims can occur on the same date. This connection allows proper time-based analysis like yearly and monthly reports.



Fraud Category

It is a calculated column classifies each claim as either "Fraud Claim" or "Genuine Claim" based on the fraud_reported field using DAX logic.

```
Fraud Category =
IF(
    'Cleaned dataset'[fraud_reported] = "Y",
    "Fraud Claim",
    "Genuine Claim"
)
```

im_amount	injury_claim	property_claim	vehicle_claim	auto_make	auto_
27700	2770	2770	22160	Toyota	Camry

Customer Segment

It is a calculated column categorizes policyholders into Basic, Standard, and Premium groups based on their annual premium amount using DAX segmentation logic.

```
Customer Segment =
SWITCH(
    TRUE(),
    'Cleaned dataset'[policy_annual_premium] < 800, "Basic",
    'Cleaned dataset'[policy_annual_premium] < 1200, "Standard",
    "Premium"
)
```

property_claim	vehicle_claim	auto_make	auto_model	auto_year	fraud
2770	22160	Toyota	Camry	2012	N
14100	63450	Accura	TI	2011	N

Risk Category

The Risk Category calculated column assigns High, Medium, or Low risk to each claim based on fraud status and claim band using conditional DAX logic.

```
Risk Category =
SWITCH(
    TRUE(),
    'Cleaned dataset'[Fraud Category] = "Fraud Claim"
        && 'Cleaned dataset'[Claim Band] = "High Claim", "High Risk",
   'Cleaned dataset'[Claim Band] = "Medium Claim", "Medium Risk",
    "Low Risk"
)
```

im	vehicle_claim	auto_make	auto_model	auto_year	fraud_reported	Incident mo

MEASURES CREATED

Total Claims

The Total Claims measure calculates the total number of insurance claims using COUNTROWS. It dynamically updates based on applied filters and slicers.
This measure provides the base volume metric for overall claim analysis.

Total Claim Amount

The Total Claim Amount measure computes the total monetary value of all claims using SUM aggregation. It reflects the overall financial exposure of the insurance company.
This measure is used in KPI cards and trend analysis visuals.

Fraud Claims

The Fraud Claims measure calculates the total number of claims marked as fraudulent using CALCULATE with a filter condition. It modifies filter context to count only records where fraud_reported = "Y". This measure helps quantify fraudulent activity in the dataset.

Fraud Rate %

The Fraud Rate % measure calculates the percentage of fraudulent claims relative to total claims using DIVIDE. It provides a key performance indicator for fraud monitoring.
The DIVIDE function ensures safe calculation by handling divide-by-zero scenarios.

Claims YTD

The Claims YTD measure calculates the total number of claims from the beginning of the year to the selected date using TOTALYTD. It enables year-to-date performance tracking.
This measure supports trend and growth analysis over time.

Claim Amount YTD

The Claim Amount YTD measure calculates cumulative claim amount from the start of the year using time intelligence functions. It helps evaluate financial performance over time.
This measure is useful for monitoring yearly claim cost trends.

Claim Amount PY

The Claim Amount PY measure calculates the total claim amount for the same period in the previous year using SAMEPERIODLASTYEAR. It allows year-over-year comparison analysis. This measure is essential for performance benchmarking and growth evaluation.

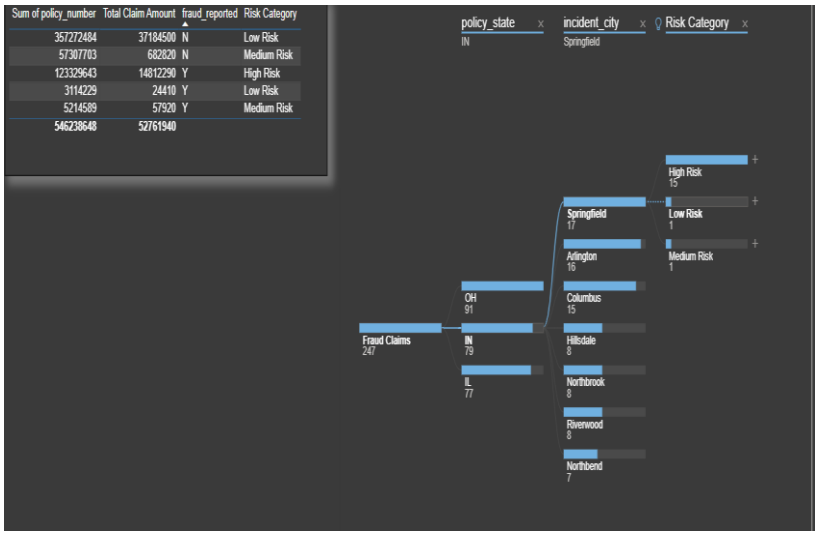
City Fraud Claims

The City Fraud Claims measure calculates the number of fraudulent claims per city using CALCULATE and filter context. It dynamically updates based on selected filters.
This measure is used for geographical fraud analysis.

City Rank

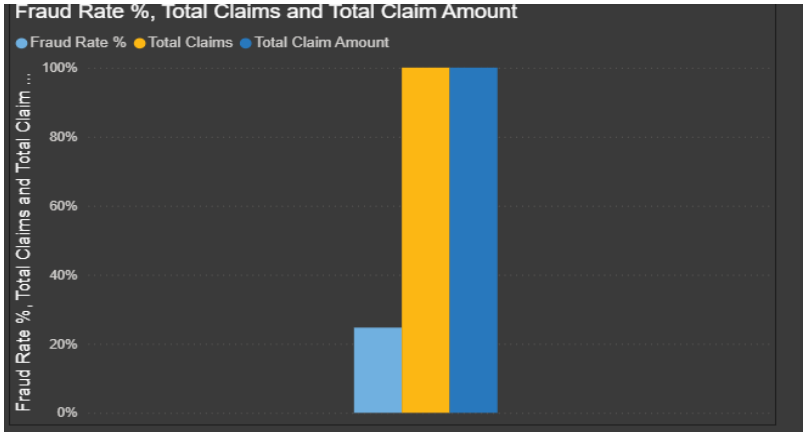
The City Rank measure ranks cities based on fraudulent claim count using RANKX.
The ALL function removes city-level filters to ensure accurate global ranking.
It is used to identify the Top 5 high-risk cities in the dashboard.

Drill-Through Page - A Drill-Through page was created to allow detailed claim-level analysis based on selected cities. Users can right-click on a city in any visual to view individual claim details such as claim number, amount, fraud status, and risk category.



Tooltip Page

A custom Tooltip page was designed to display additional KPIs when hovering over visuals.
It enhances user experience by providing quick summary insights without navigating away from the main dashboard.



Insights & Key Findings

Fraud Concentration in Specific States

Fraud rate analysis revealed that certain states show a significantly higher percentage of fraudulent claims compared to others. This indicates geographical concentration of fraud risk and highlights regions that require stricter monitoring and investigation.

High Claim Amounts Have Higher Fraud Risk

Claims categorized under the High Claim Band showed a higher proportion of fraud cases. This suggests that larger claim amounts are more likely to be associated with fraudulent activity, increasing financial exposure.

High-Risk Category Dominated by Fraudulent High Claims

The Risk Category analysis identified that most High-Risk claims are linked to fraudulent and high-value claims. This confirms that combining claim amount and fraud status provides an effective risk classification approach.

Certain Cities Show Higher Fraud Frequency

The Top 5 Cities ranking identified specific cities with significantly higher fraud claim counts. These cities may require enhanced fraud detection mechanisms and policy review.

Demographic Influence on Fraud Patterns

Age Band analysis indicated variations in fraud distribution across demographic segments. Certain age groups showed relatively higher fraud proportions, suggesting possible behavioral or risk-based patterns.

Customer Segment Impact

Premium and higher annual premium segments contributed more to total claim amounts. However, fraud distribution varied across segments, indicating that risk is not solely dependent on premium value.

Time-Based Trends Support Monitoring

Using time intelligence (YTD and previous year comparison), claim trends and fraud variations over time were identified. This supports proactive fraud detection and performance monitoring.

Dashboard Overview

The dashboard was designed to provide a comprehensive view of insurance claim patterns, fraud trends, and risk segmentation. It combines KPI indicators, geographical analysis, risk classification, and ranking insights into a single interactive report page.

KPI Summary Section - The top section contains key performance indicators including Total Claims, Total Claim Amount, Fraud Claims, Fraud Rate %, and Average Claim

Amount. These KPIs provide a high-level overview of overall claim volume, financial exposure, and fraud proportion in the dataset.

Fraud Rate by State -This visual analyzes fraud percentage across different states. It helps identify high-risk regions and supports geographical fraud pattern detection.

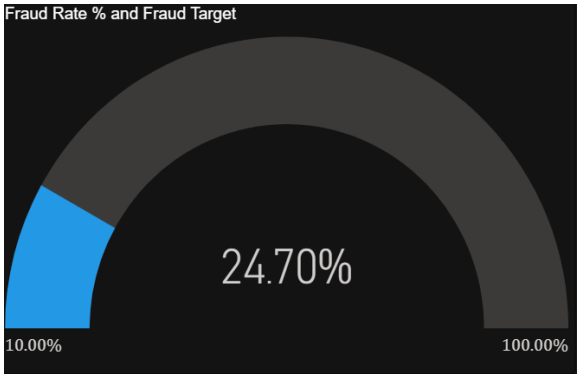
Claim Band Distribution - This chart displays total claim amount segmented into Low, Medium, and High claim bands. It helps assess financial exposure across different claim size categories.

Risk Category Analysis - This visual classifies claims into High, Medium, and Low risk categories. It supports risk profiling and helps management focus on high-risk segments.

Top 5 Cities by Fraud Claims -This ranking visual identifies the top five cities with the highest number of fraudulent claims using RANKX. It supports targeted fraud investigation and resource allocation.

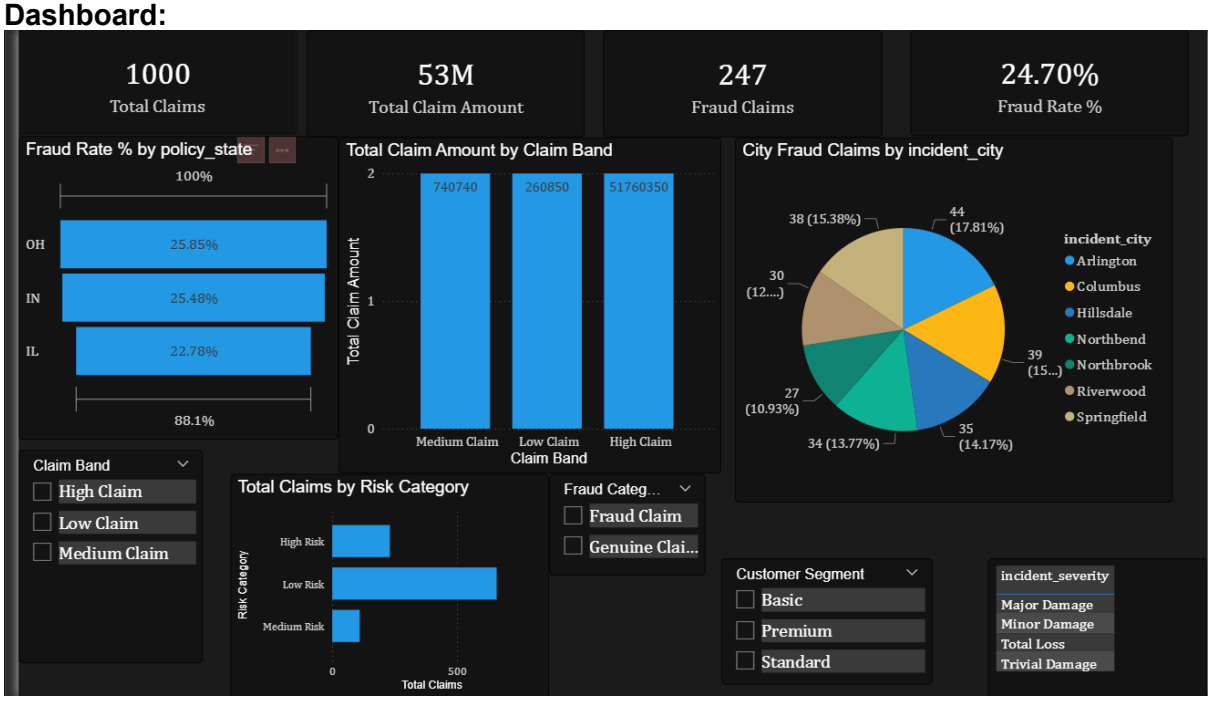
Fraud Rate vs Target

A Gauge visual was created to display the Fraud Rate percentage against a predefined target value. The minimum was set to 0 and the maximum to 25% to provide a realistic performance range. This visualization enables quick comparison of actual fraud levels against the expected benchmark.



Interactive Slicers Slicers were added for Fraud Category, Claim Band, Region Group, Customer Segment, and Incident Severity. These slicers allow dynamic filtering across all visuals, enhancing dashboard interactivity and user-driven analysis.

Dashboard:



Conclusion:

The dashboard demonstrates that fraud risk is multidimensional, influenced by claim amount, geography, and customer characteristics. By integrating star schema modeling, DAX-based segmentation, ranking logic, and time intelligence, the solution provides a scalable analytical framework for proactive fraud detection and operational risk management. The interactive reporting environment enables data-driven decision-making at both executive and operational levels.